System Architecture: Bayesian Signal Convergence Engine

PRIMARY SIGNAL LAYERS

GEO **Geopolitical Events**
Sources: ACLED conflict data, GDELT event feeds, curated news feeds

MKT **Market Indicators**
Sources: Alpha Vantage, FRED economic data, exchange data feeds

OSI **Open Source Intelligence**
Sources: OSINT feeds, aircraft tracking (ADS-B), social signals

SYS **Systemic Risk**
Sources: Volatility indices, contagion metrics, correlation breakdowns

Confidence-Weighted Signals

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Bayesian Convergence Engine

- Sequential Bayesian Updating
- Regime-Aware Adaptive Weighting
- Cross-Layer Correlation Detection
- Multi-Source Signal Fusion
- Anomaly & Threshold Detection

↑ Context Input

Narrative Context Layer
Calendar/Celestial actor-belief context. Max 0.5 bonus modifier. No convergence weight. Provides cultural and temporal framing for actor behaviour modelling.

↻ **Prediction Calibration Feedback Loop (Brier Scoring) feeds back into Adaptive Weighting**

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INTELLIGENCE OUTPUTS

Composite Risk Assessments
Multi-factor risk scoring with confidence intervals

AI-Generated Intelligence Briefs
Automated thesis generation with source attribution

Game Theory Scenarios
Actor strategy modelling, escalation trajectories

Prediction Tracking
Accuracy scoring with Brier calibration metrics

Institutional Delivery Layer

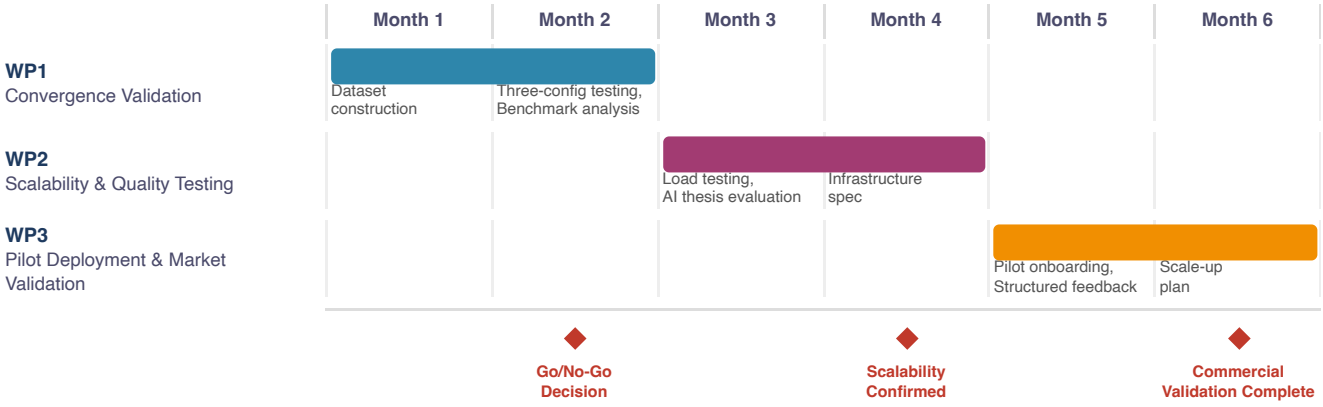
RESTful API | Interactive Dashboard | AI Chat Analyst Interface | War Room (Geospatial) | Alerting System

Validation Benchmark Matrix

Event Category	Example Events	Signal Layers Active	Baseline (Single-Source) Brier Score	Target (Converged) Brier Score	Expected Lead Time Improvement
Military Escalation	Russia-Ukraine 2022, Israel-Gaza 2023	GEO + OSI + MKT + SYS	0.45	< 0.30	**+24-48 hours**
Sanctions & Trade	Russia sanctions, China tech restrictions	GEO + MKT + SYS	0.40	< 0.28	**+12-36 hours**
Market Disruption	SVB collapse, Energy price shocks	MKT + SYS + GEO	0.42	< 0.30	**+6-24 hours**
Political Instability	Coups, elections, regime change	GEO + OSI	0.50	< 0.35	**+48-72 hours**
Systemic Risk Events	Contagion, correlation breakdown	SYS + MKT	0.48	< 0.32	**+12-48 hours**

Brier scores range 0 (perfect) to 1 (worst). Scores below 0.25 indicate strong predictive calibration. Baseline estimates derived from single-source analytical benchmarks. Targets to be validated in Work Package 1.

Project Timeline



Validation Dataset: 200+ historical geopolitical-market events (2020-2025) | **Statistical Target:** $p < 0.05$ improvement over baselines | **Pilot Users:** 3-5 institutional organisations